OPSINTEL AI

Evidence-based pipeline investigation

From failed overnight run to traceable next action - powered by Exasol Personal

EXATHON INDIA 2026 / TRACK 04: THE WORK BEHIND THE WORK

Public prototype • MIT licensed • Synthetic data

THE PROBLEM

The failure is visible. The explanation is scattered.

WHO

On-call data engineers

They must reconcile run status, expected rows, actual rows, and logs while the business waits for trusted data.

TODAY

01 Find the failed run

02 Compare source and target counts

03 Search logs across systems

04 Decide what to inspect next

Result: slow first diagnosis and weak evidence traceability

PRODUCT WORKFLOW

One path from operational evidence to a safe next action

01 **Pipeline runs and logs** →

02 **Exasol analytical features** →

03 **Evidence classification** →

04 **Candidate cause and next action** →

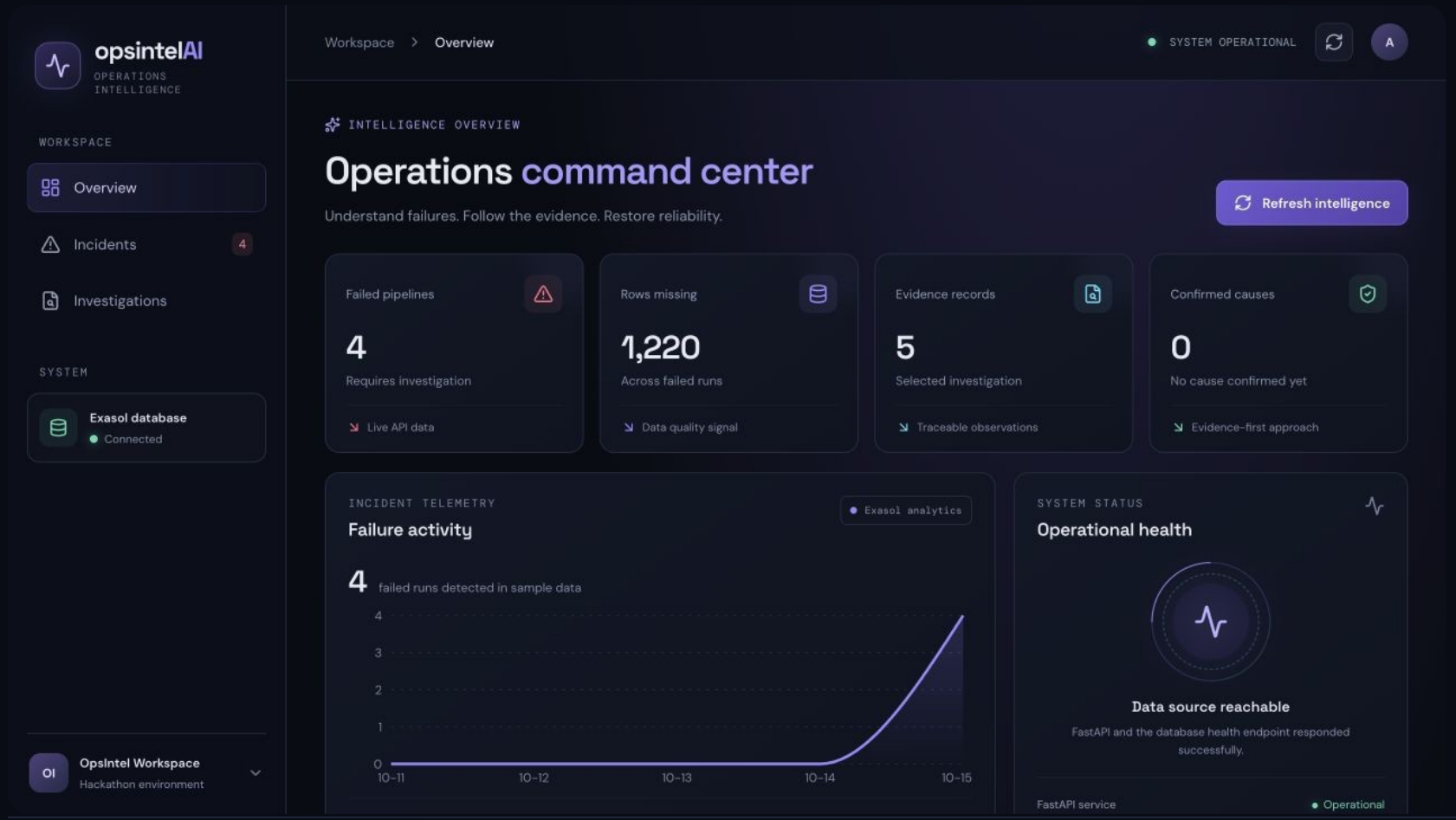
05 **Engineer verifies and confirms**

Safety boundary

Database observations are facts. Explanations remain candidates. No repair runs automatically.

LIVE PRODUCT

Operational telemetry comes from Exasol, not a mock chart



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failed pipeline runs

1,220

missing rows surfaced

✓ Exasol connected
✓ Daily trend calculated
✓ Incidents selectable

INVESTIGATION EXPERIENCE

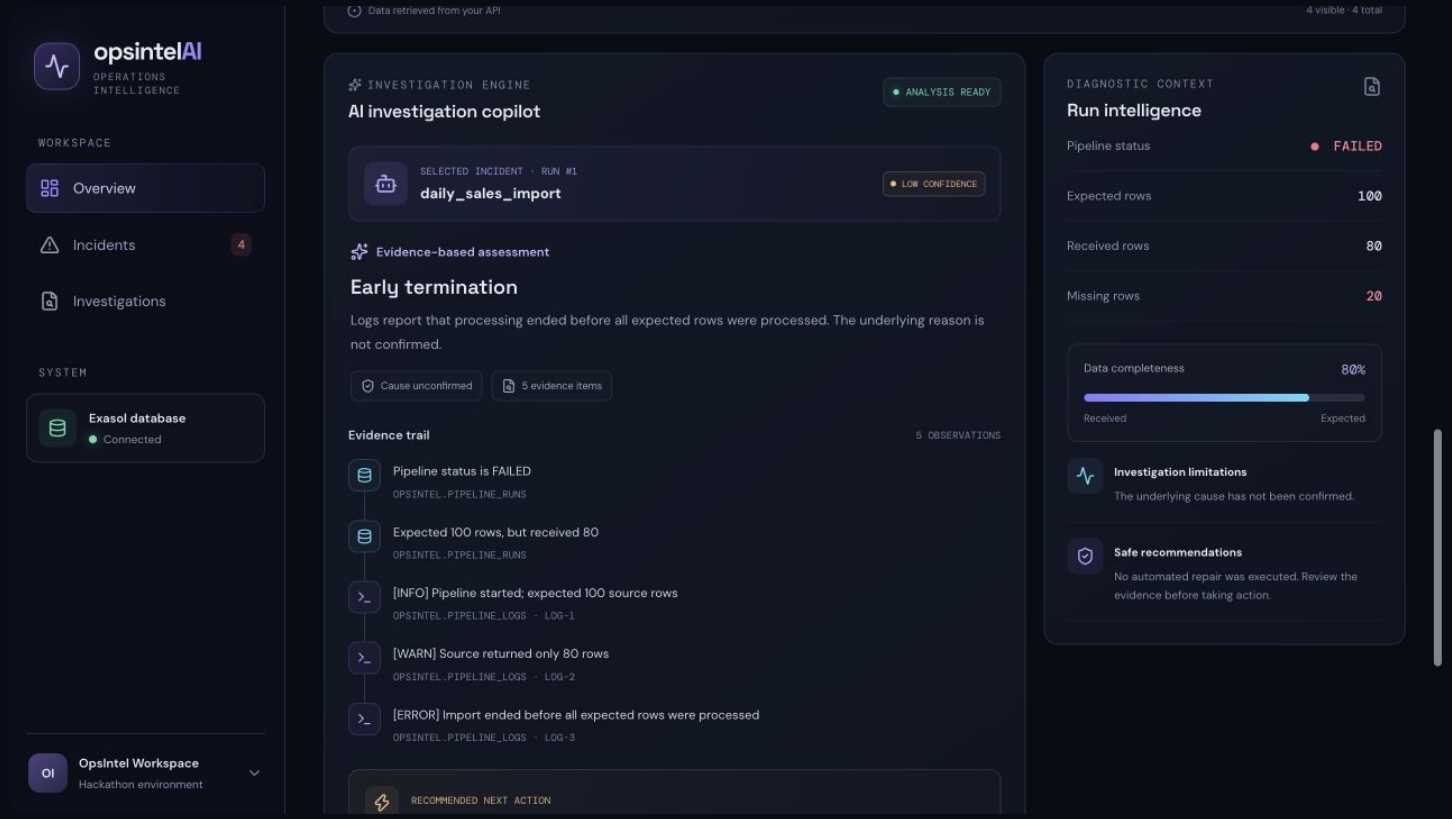
Every recommendation points back to source evidence

Selected run

daily_sales_import

Expected 100 rows
Received 80 rows
Missing 20 rows

Five observations identify the exact Exasol table and log record used.



Candidate: early termination • confidence: low

WHY EXASOL

The database calculates operational context before the API responds

ANALYTICAL FEATURES

01 Previous run status with LAG

02 Rolling actual-row baseline

03 Rolling failure rate

04 Missing rows and completeness

05 Daily incident telemetry

WINDOW FUNCTION

```
AVG(failed) OVER (
  PARTITION BY pipeline_name
  ORDER BY run_time
  ROWS BETWEEN 19 PRECEDING
  AND CURRENT ROW
) * 100
```

Thin API → reusable SQL evidence → consistent dashboard

MEASURED EVIDENCE

One million runs stay below 0.51 seconds at p95 locally



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predefined synthetic challenge cases

3.025 s

one-million-row import

473 ms

median analytical query

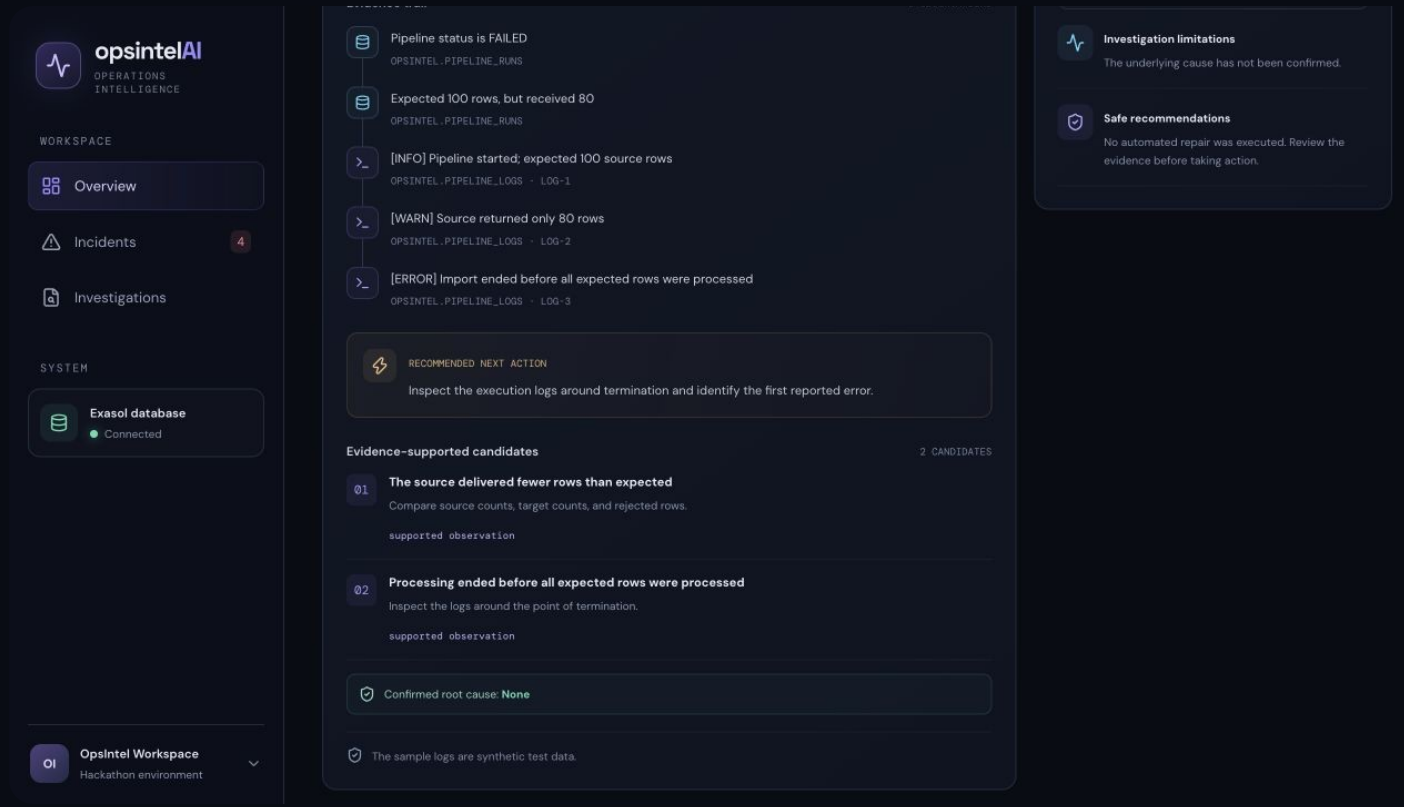
45/45

automated tests passed

Source: reports/exasol-benchmark.json • Local single-user run on 18 Oct 2026 • Not a concurrency claim

TRUST AND SAFETY

The product is designed to abstain before it overclaims



OBSERVATION

Source-backed facts

CANDIDATE

Evidence-supported possibility

CONFIRMATION

Reserved for human verification

No automatic repair • confirmed root cause remains null

VALIDATION ROADMAP

What is proven, and what must be proven next

PROVEN NOW

01 Live Exasol-backed demo

02 One-million-row benchmark

03 45 automated tests

04 Traceable candidates and abstention

READY TO RUN

01 Reviewer-controlled holdout

02 Frozen-commit evaluation

03 Human impact study protocol

04 Clean-clone rehearsal

PRODUCTION WORK

01 Orchestrator connectors

02 Authentication and authorization

03 Concurrent workload testing

04 Audited retention and monitoring

The repository does not claim independent accuracy or measured human productivity before those studies happen.

OPSINTEL AI

Follow the evidence. Investigate faster. Confirm carefully.

github.com/Anafa-thabassum/opsintel-ai

Live demo: one incident, five observations, one safe next action